

Kirin Feng

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Summary

Cybersecurity graduate with hands-on experience in application security, digital forensics, threat modeling, vulnerability assessment, and secure software development. Applied OWASP SAMM, BSIMM, STRIDE, OWASP ZAP, Autopsy, and secure SDLC practices across graduate-level security projects involving multi-tenant SaaS architecture, web application testing, forensic investigation, and API security. Experienced in analyzing attack surfaces, identifying security gaps, validating digital evidence, and translating technical findings into practical remediation recommendations. Brings additional experience in software development, databases, Linux systems, and client-facing technical work, with a strong foundation for Security Analyst, Application Security, and Junior Security Engineer roles.

Technical Skills

Security & Assessment: Nmap, Burp Suite, WebGoat, Penetration testing, OWASP ZAP, DVWA, WebGoat, Wireshark, Autopsy, IDA Freeware, Angry IP Scanner, vulnerability assessment, web application security, HTTP interception and analysis

Security Engineering: STRIDE, data flow diagrams, trust boundaries, OWASP SAMM, BSIMM, secure SDLC, threat modeling, API security, authentication and authorization, tenant isolation, risk assessment

Digital Forensics: Autopsy, E01 disk images, Windows artifacts, browser history analysis, filesystem analysis, evidence integrity, MD5/SHA hash verification, forensic reporting; academic exposure to Volatility, Scalpel, Guymager, dd, dc3dd, and file carving

Systems & Networking: Linux, Windows, Kali Linux, TCP/IP, Cisco networking, VirtualBox, Cisco Packet Tracer, PowerShell, network troubleshooting

Programming languages: Python, Java, JavaScript, C++, PHP, SQL, HTML, CSS, Assembly language, Visual Basic, Classic Visual Basic

Databases & Tools: MySQL, PostgreSQL, Microsoft Access, Tableau, MATLAB, GitHub, Visual Studio, Eclipse, Processing, Microsoft Project, StarUML, Remix, MetaMask, Ethereum Sepolia testnet

Relevant Experience

Website Developer –January 2023 to August 2023

New Field Education Group – Vancouver, BC

- Supported the development and maintenance of the website functionality and user-facing content using HTML, CSS, JavaScript, and related web technologies.
- Translated business and user requirements into practical website updates, coordinated feedback with stakeholders, and supported testing and issue resolution.
- Resolved website issues and improved usability and consistency across desktop and mobile platforms.
- Presentation through structured page layouts, consistent styling, and iterative quality assurance on different platforms.

Business Development Advisor –January 2021 to August 2021

Quantasphere Solution – Burnaby, BC

- Supported business development planning, client communication, and opportunity evaluation by organizing requirements and preparing practical recommendations.
- Coordinated with stakeholders to clarify priorities, track deliverable, and align technical or service activities with business objectives.
- Contributed research, documentation, and presentation support for decision-making and relationship development.

Cybersecurity & Technical Projects

Authorized Web Application Penetration Test – Graduate Industry Partner Project

Nmap | Nessus Essentials | Burp Suite | DIRB | JWT Security Testing | Python | Web Application Security

- Conducted an authorized external penetration test on two web application environments, building an asset, service, and version baseline while mapping the attack surface through network, web, and client-side analysis.
- Used Nmap, Nessus Essentials, Burp Suite, DIRB, JWT analysis, and Python-based source-map review to validate findings across authentication, access control, exposed services, security headers, file upload handling, and dependency risk.
- Reconstructed 76 client-side source files from an exposed Webpack source map to assess source code disclosure, route and API exposure, and business logic leakage risk.
- Delivered prioritized remediation guidance for JWT signature verification, server-side authorization, source-map restriction, dependency patching, firewall rules, secure upload controls, and HTTP security headers.

Software Security Assessment & Threat Modeling – Graduate Academic Project

OWASP SAMM | BSIMM | STRIDE | DFD | Microservices | Secure SDLC

- Assessed the software security maturity of a simulated multi-tenant e-commerce SaaS organization using OWASP SAMM and a BSIMM quick survey.
- Identified material gaps in governance, secure build practices, defect management, and incident management.
- Designed data flow diagrams for 7 major system components: a customer SPA, merchant dashboard, API gateway, authentication, order, checkout, and payment services and explicit trust boundaries across microservices, managed data stores, and third-party integration.
- Prioritized risks using severity and likelihood assessments to analyze spoofing, tampering, repudiation, information disclosure, denial-of-service, and privilege-escalation scenarios, with emphasis on API access control and tenant isolation.
- Developed a prioritized security roadmap covering standardized authentication and authorization, CI/CD security checks, monitoring, vulnerability testing, governance, and incident response; used OWASP Cornucopia to reinforce secure-design analysis.

Digital Forensics Investigation Using Autopsy – Graduate Academic Project

Autopsy | E01 | Windows Forensics | Hash Verification | Evidence Reporting

- Conducted a structured investigation of a Windows E01 forensic image in Autopsy and validated image integrity using available cryptographic acquisition hashes.
- Examined Windows user-account artifacts, browser history, filesystem records, and user-generated files to reconstruct activity and answer defined investigative questions.
- Collected, correlated, and documented digital evidence in a repeatable workflow, preserving evidence context.
- Produced a structured forensic report documenting evidence and investigation findings.

Web Application Vulnerability Assessment – Graduate Security Project

OWASP ZAP | DVWA | WebGoat | Kali Linux | HTTP Proxy

- Performed authorized security testing against intentionally vulnerable DVWA and WebGoat environments using OWASP ZAP.
- Configured an intercepting proxy and HTTP breakpoints to inspect and modify login and registration requests, parameters, and headers; investigated malformed requests and HTTP 400/404 responses.
- Executed multiple automated scans, reviewed alerts and risk ratings, analyzed request-response behavior, and generated HTML security reports with remediation considerations.

Ethereum Marketplace DApp – Graduate Blockchain Project

Solidity | Ethereum Sepolia | Remix | MetaMask | JavaScript

- Developed and deployed a decentralized marketplace smart contract to the Ethereum Sepolia testnet using Solidity and Remix.
- Implemented item listing, purchase, seller cancellation, status transitions, require-based transaction validation, view functions, and on-chain event logging.
- Integrated MetaMask for wallet-based interactions and verified deployment and transaction activity through contract addresses, hashes, and blockchain explorer records.

- Verified smart contract transactions through Sepolia Explorer.

Blockchain Architecture & Security Analysis – Graduate Research Project

Blockchain Architecture | Consensus | Tokenomics | Trust Models | Security Risk

- Researched a blockchain ecosystem's architecture, design, consensus assumptions, tokenomics, and trust model, evaluating attack surfaces, protocol risks, and decentralization trade-offs.
- Synthesized technical findings into an academic presentation, an 8-10 minute briefing, and a concise written brief supported by references.

PIPEDA Privacy & Security Case Analysis – Graduate Academic Analysis

PIPEDA | Privacy | Consent | Data Protection | Regulatory Analysis

- Analyzed Canadian privacy and security cases involving consent, hashed identifiers, data use, and organizational accountability under PIPEDA and Office of the Privacy Commissioner findings.
- Translated regulatory issues into practical privacy, governance, and risk considerations relevant to security and compliance functions.

Additional Engineering Projects

Raspberry Pi Surveillance Robot – Undergraduate Team Project

Project Leader/software developer

Raspberry Pi | Linux | Computer Vision | Face Detection | Mobile Control

- Designed and implemented software modules for a Raspberry Pi surveillance robot based on project requirements.
- Implemented robot moving, obstacle avoidance, line tracking, text to speech, color detection, face detection and mobile app control.
- Prepared technical documentation covering software requirements, assembly, testing, deployment, and maintenance.

Smart Health System – Undergraduate Software Project

Chief Front-End Developer

Front-End Development | UI Design | Software Testing

- According to project requirements and designed the appropriate user interface.
- Implemented user requested features based on functional requirements.
- Developed and tested all features and successfully made the software work.

Bookstore System – Undergraduate Software Project

UI/UX Designer

Application Development | Database Design | Systems Analysis

- Developed user interface components and optimized user workflows across multiple platforms.
- Implemented the application on different platforms.

Facial Recognition System – Undergraduate Team Project

Project Leader/software developer

Raspberry Pi | Linux | Facial Recognition

- Led a three member team on the successful development of facial recognition system on raspberry pi using the built-in Linux system.
- Moderated different ideas between members to achieve the goal.
- Created documentation of software requirements, assembling and installing, testing, as well as implementation for the project.

Education

Master of Science: Cybersecurity – September 2024 to May 2026

New York Institute of Technology – Vancouver, BC

- Relevant study: software security, digital forensics, web application security, network security, blockchain technologies, privacy, risk, and security governance.

Bachelor of Technology: Information Technology – *May 2018 to August 2024*

Kwantlen Polytechnic University – Surrey, BC

- Foundation in software development, web technologies, Data Mining, Database Management, systems analysis, UI/UX, networking, and team-based project delivery.

Research & Professional Development

AI Bias in Hiring Systems - *Graduate Research and Presentation*

- Researched bias in AI-powered hiring processes and communicated findings through a referenced academic poster and oral presentation.

Certifications

CCNA: Introduction to Networks (ITN) – August 2024

Cisco Network Academy